

LOAD BALANCER

Importance of load balancer:

- 1 Distributes incoming traffic evenly across multiple servers to prevent overload.
- 2 Ensures high availability by routing traffic only to healthy instances.
- 3 Increases fault tolerance by automatically handling instance or AZ failures.
- 4 Improves application performance and reduces latency.
- 5 Works with Auto Scaling to handle traffic spikes smoothly.
- 6 Enhances security with SSL termination, WAF, and DDoS protection.
- 7 Provides centralized monitoring and logging for better visibility and troubleshooting.

Tools going to be used for the task:

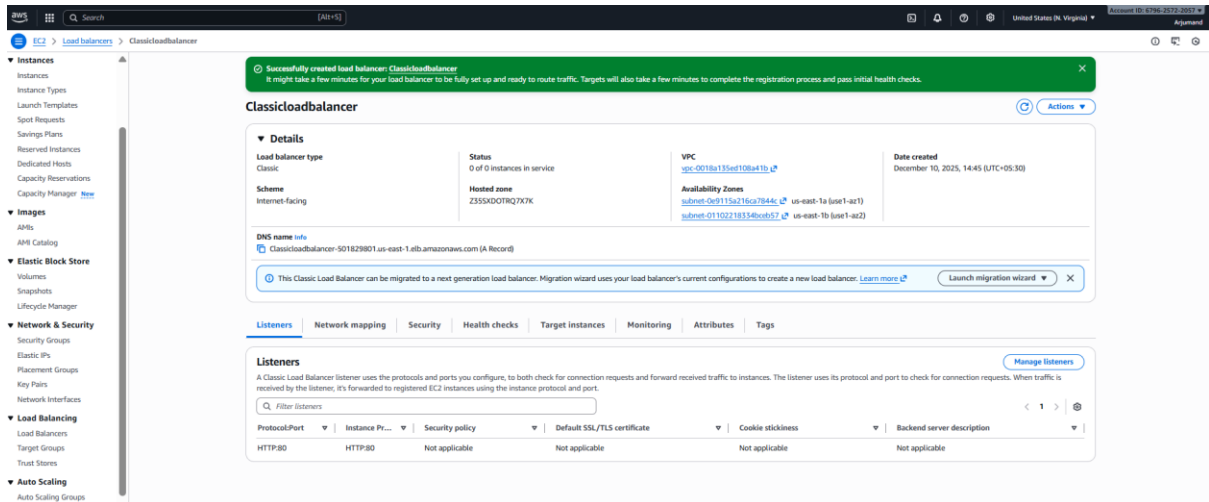
1. **EC2** → to launch instances for all load balancers.
2. **VPC (subnets + security groups)** → networking setup for CLB, ALB, NLB.
3. **Target Groups** → required for ALB and NLB routing.
4. **ACM (AWS Certificate Manager)** → to generate and attach SSL certificate to ALB.
5. **Route 53** → to map ALB DNS using an Alias record.
6. **S3 Bucket** → to store Application Load Balancer access logs.

1. Configure Classic Load balancer.

Steps:

- Go to EC2 → Load Balancers → Create Load Balancer → Classic Load Balancer.
- Choose Internet-facing, select public subnets, and keep listener HTTP:80.
- Configure security groups to allow inbound HTTP (port 80).
- Add health check using HTTP on path / and default thresholds.
- Register your EC2 instances, save, and verify they show InService.

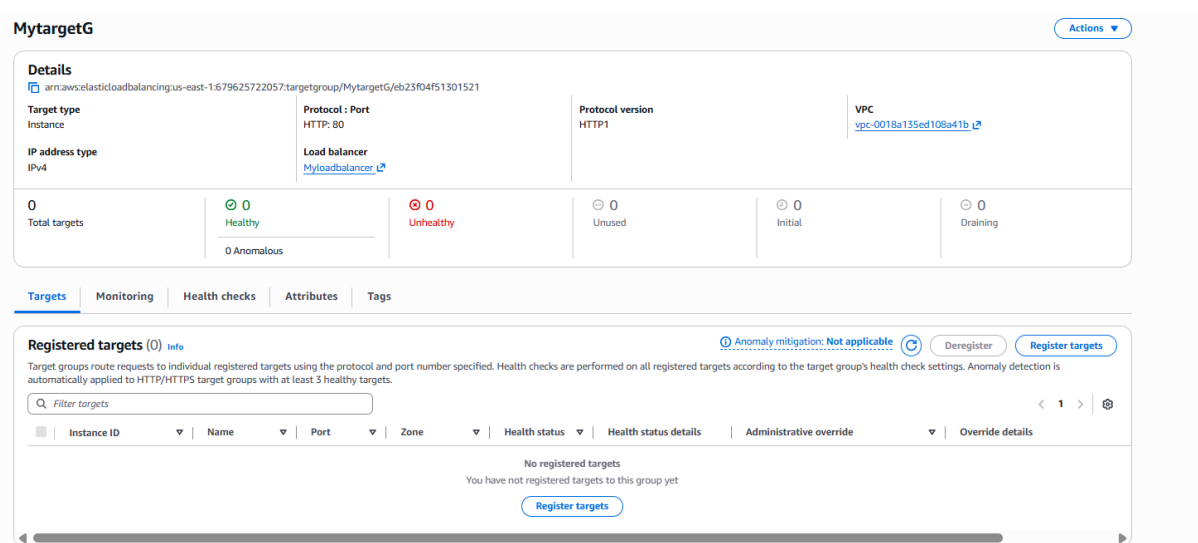
- I have configured classic load balancer with two public subnets in different zones



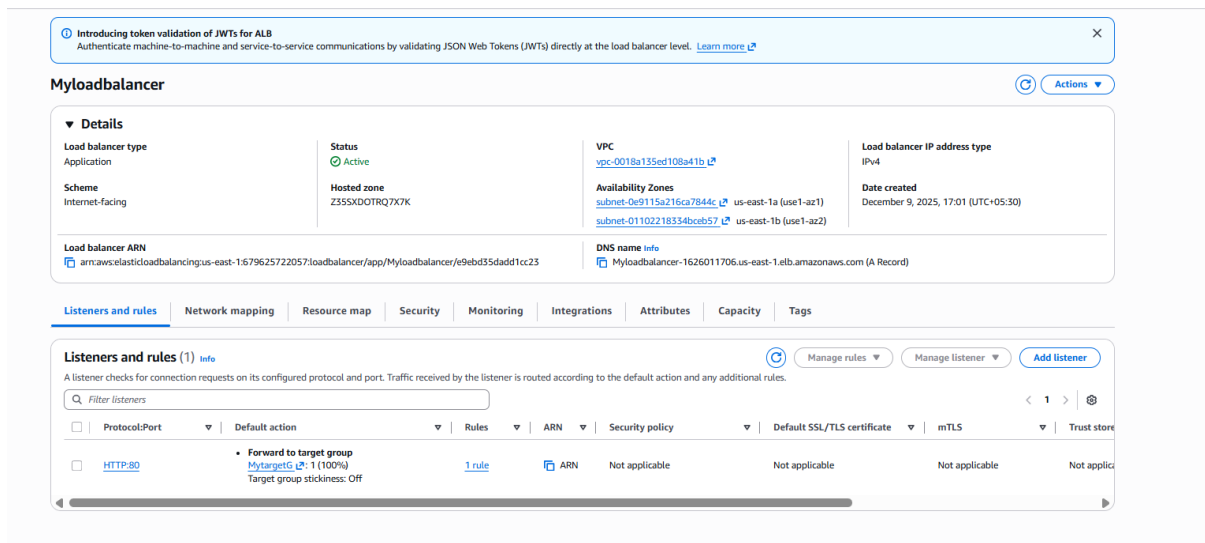
2. Configure Application Load balancer.

Steps:

- Go to **EC2 → Load Balancers → Create Load Balancer → Application Load Balancer**.
- Select **Internet-facing**, choose **IPv4**, and pick **at least two public subnets**.
- Create/choose a **Security Group** that allows inbound HTTP (80).
- Create a **Target Group** (Instance type → HTTP:80) and **register your EC2 instances**.
- In the ALB Listener section, add listener **HTTP:80 → Forward to Target Group**.
- Create the ALB and verify that **Targets show “healthy”** and ALB DNS loads your website.



- Created ALB using two public subnets and one target group and security group



3. Configure Network Load balancer.

Steps:

- Go to EC2 → Load Balancers → Create Load Balancer → Network Load Balancer.
- Enter a name, choose Internet-facing, and keep TCP:80 as the listener.
- Select two public subnets across different Availability Zones.
- Create a new Target Group with Target Type = Instance and Protocol = TCP:80.
- Register your EC2 instances as targets and save the target group.
- Attach the target group to the NLB and click Create Load Balancer.
- Test using the NLB DNS name to confirm traffic reaches your backend EC2 instances.

Target type
Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ **Instances**
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
Suitable for: [ALB](#) [NLB](#) [GWLB](#)

☐ **IP addresses**
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: [ALB](#) [NLB](#) [GWLB](#)

☐ **Lambda function**
Supports load balancing to a single Lambda function. ALB required as traffic source.
Suitable for: [ALB](#)

☐ **Application Load Balancer**
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: [NLB](#)

Target group name
Name must be unique per Region per AWS account.

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 0/32

Protocol
Protocol for communication between the load balancer and targets.

Port
Port number where targets receive traffic. Can be overridden for individual targets during registration.

1-65535

IP address type
Only targets with the indicated IP address type can be registered to this target group.

☒ **IPv4**
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ **IPv6**
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

[Create VPC](#)

EC2 > **Target groups** > Create target group

Step 1

Create target group

Step 2 - recommended

Register targets

Step 3

Review and create

Review and create
Review your target group configuration before creating

Step 1: Target group details

Target group details			
Name	Target type	Protocol - Port	Protocol version
NLBtarget	Instance	TCP: 80	HTTP1
VPC	IP address type		
vpc-0018a135ed108a41b	IPv4		

Health check details

Health check protocol	Health check path	Health check port	Interval
HTTP	/	traffic-port	30 seconds
Timeout	Healthy threshold	Unhealthy threshold	Success codes
6 seconds	5	2	200-599

Step 2: Register targets

Targets (0)			
Instance ID	Name	Port	Zone
No targets added			

[Cancel](#) [Previous](#) [Create target group](#)

Created a new target group for NLB

Review
Review the load balancer configurations and make changes if needed. After you finish reviewing the configurations, choose **Create load balancer**.

Summary
Review and confirm your configurations. [Estimate cost](#)

Basic configuration Edit Name: mynlb Scheme: Internet-facing IP address type: IPv4	Network mapping Edit VPC: vpc-0018a135ed108a41b Availability Zones and subnets: - us-east-1a subnet-023fb7044fd931fb8 p1-subnet-N1 - us-east-1b subnet-01102218334bceb57 mysubnet2	Security groups Edit default sg-09a5507d74a26c8	Listeners and routing Edit TCP:80 Forward to 1 target group
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Service integrations [Edit](#)
AWS Global Accelerator: -

Tags [Edit](#)
-

Attributes

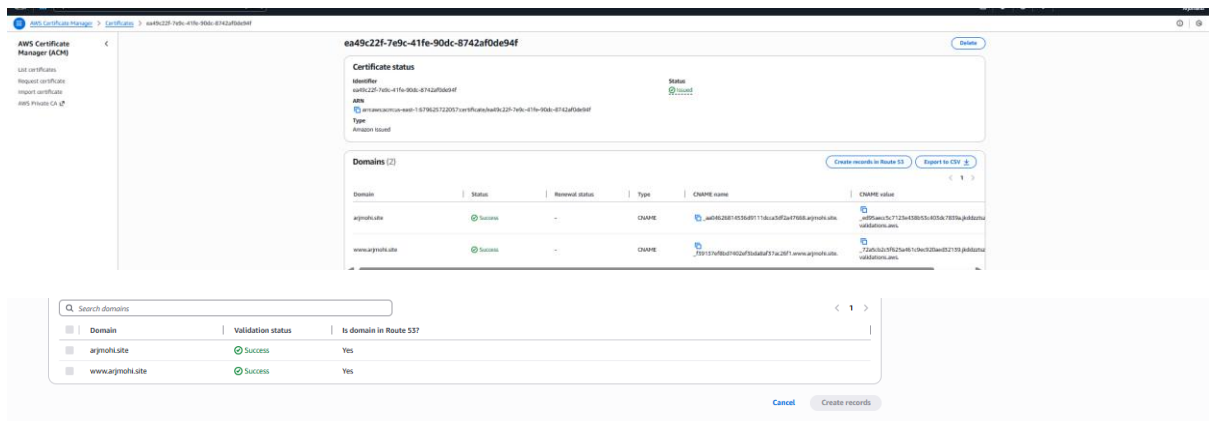
Creation workflow and status

► **Server-side tasks and status**
After completing and submitting the above steps, all server-side tasks and their statuses become available for monitoring.

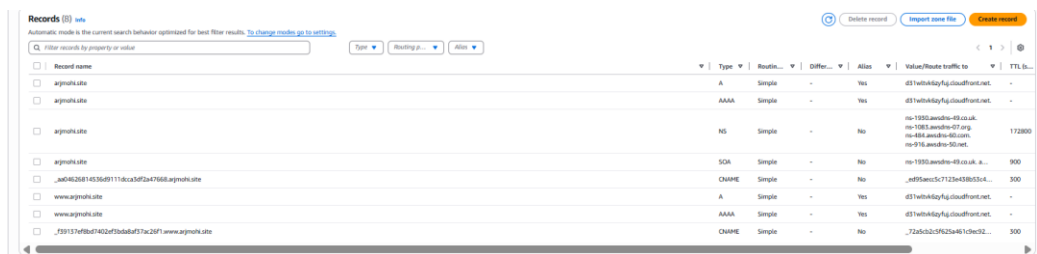
[Cancel](#) [Create load balancer](#)

NLB created by new target group and two subnets

- We can see the records after making certificate with my domain



- R53 records as we can see related to my domain



- Added ssl to my APLB and added certificate source

Pre-routing action | Info

☒ No pre-routing action

☐ Authenticate user
User auth with OIDC or Amazon Cognito

☐ Validate token
Validate client-presented JSON Web Token (JWT)

Routing action

☒ Forward to target groups

☐ Redirect to URL

☐ Return fixed response

Forward to target group | Info
Choose a target group and specify routing weight or [create target group](#).

Target group
MytargetG
Target type: TCP, UDP, HTTP, HTTPS, gRPC, WebSocket, or Custom
Target stickiness: Off

Weight
1
0-999

Percent
100%

Target group stickiness | Info
Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

☐ Turn on target group stickiness

Secure listener settings | Info

Security policy | Info
Your load balancer uses a Secure Sockets Layer (SSL) negotiation configuration called a security policy to manage SSL connections with clients. [Compare security policies](#).

Security category
All security policies

Policy name | [New](#)
ELBSecurityPolicy-TLS13-1-2-Res-PQ-2025-09 (Recommended)

Default SSL/TLS server certificate
The certificate used if a client connects without SNI protocol, or if there are no matching certificates. You can source this certificate from AWS Certificate Manager (ACM), Amazon Identity and Access Management (IAM), or import a certificate. This certificate will automatically be added to your listener certificate list.

Certificate source

☒ From ACM

☐ From IAM

☐ Import certificate

Certificate (from ACM)
The selected certificate will be applied as the default SSL/TLS server certificate for this load balancer's secure listeners.

arjnohi.site
b67950a8-acae-47ef-bd5b-559662d805c

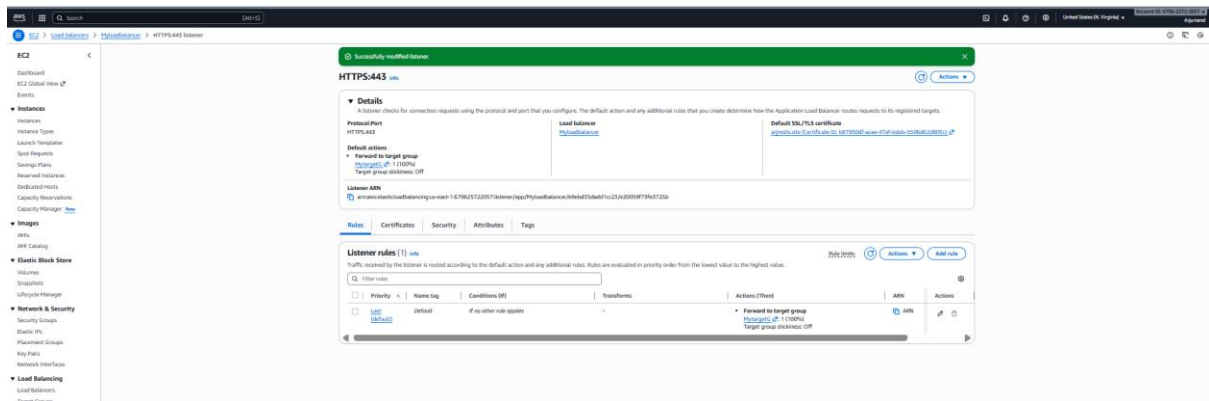
[Request new ACM certificate](#)

Client certificate handling | Info
Client certificates are used to make authenticated requests to remote servers. [Learn more](#).

☐ **Mutual authentication (mTLS)**
Mutual TLS (Transport Layer Security) authentication offers two-way peer authentication. It adds a layer of security over TLS and allows your services to verify the client that's making the connection.

Server-side tasks and status
After completing and submitting the above steps, all server-side tasks and their statuses become available for monitoring.

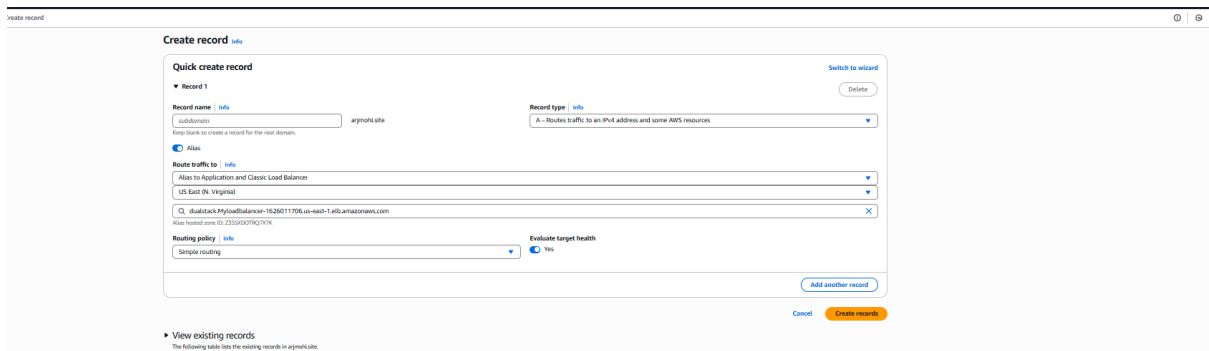
[Cancel](#) [Save changes](#)



5. Map Application load balancer to R53.

Steps:

- Open Route53 hosted zone for your domain.
 - Create an **A record** with Alias = **Yes**.
 - Select “Alias to Application Load Balancer.”
 - Choose your ALB from the drop-down list (dualstack).
 - Save the record and verify that the domain resolves to ALB.
- Mapped application load balancer to route 53



- New records are not added old records are deleted now records redirecting my load balancer

Hosted zone details Edit hosted zone

Records (6) Accelerated recovery DNSSEC signing Hosted zone tags (0)

Records (6) [Auto](#)

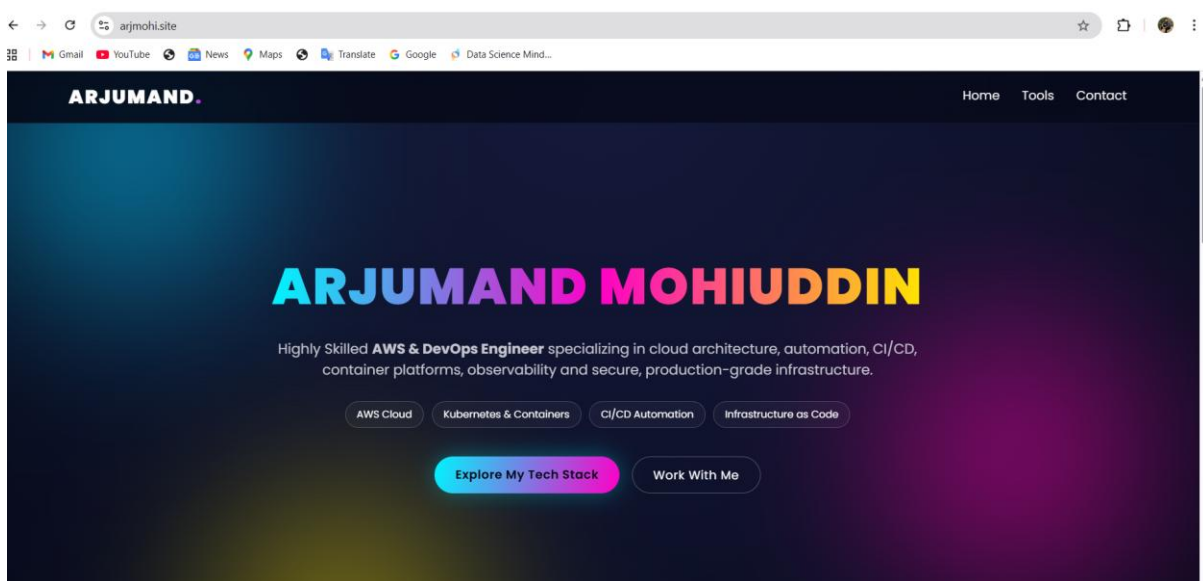
Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value

Type Routing p... Alias

Record name	Type	Routing...	Differ...	Alias	Value/Route traffic to	TTL (s)
arjmohi.site	A	Simple	-	Yes	duatstack.myloadbalancer-1...	-
arjmohi.site	AAAA	Simple	-	Yes	d57b6b4d4yfg.cloudfront.net	-
arjmohi.site	NS	Simple	-	No	ns-1930.awdns-49.co.uk. ns-1930.awdns-47.org. ns-484.awdns-40.com. ns-918.awdns-50.net.	172800
arjmohi.site	SOA	Simple	-	No	ns-1930.awdns-49.co.uk. a...	900
_uad4626814536d9111db.ukbf2a4766d.arjmohi.site	CNAME	Simple	-	No	_n895awnc5c7125a458b53d...	300
www.arjmohi.site	A	Simple	-	Yes	duatstack.myloadbalancer-1...	-

- This is my website is deployed after the new records



6. Push the application load balancer logs to S3.

Steps

- Created an S3 bucket in the same region as the ALB to store access logs.
- Added a bucket policy allowing logdelivery.elb.amazonaws.com to write logs.
- Opened ALB in EC2 → Load Balancers → Attributes tab.
- Enabled "Access Logging" and selected the S3 bucket.
- Added optional prefix for log organization.
- Saved configuration to start log delivery.
- Verified log files in S3 after 5–10 minutes.

- Created bucket then added bucket policy to receive load balancer logs

bucketforloadbalancer1 [Info](#)

Objects | Metadata | Properties | **Permissions** | Metrics | Management | Access Points

Permissions overview

Access finding
Access findings are provided by IAM external access analyzers. Learn more about [How IAM analyzer findings work](#).
[View analyzer for us-east-1](#)

Block public access (bucket settings) [Edit](#)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

Block all public access
Off
► **Individual Block Public Access settings for this bucket**

Bucket policy [Edit](#) [Delete](#)

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AWSALBLogging",
      "Effect": "Allow",
      "Principal": {
        "Service": "logdelivery.elasticloadbalancing.amazonaws.com"
      },
      "Action": "s3:PutObject",
      "Resource": "arn:aws:s3:::bucketforloadbalancer1/*",
      "Condition": {
        "StringEquals": {
          "s3:x-amz-acl": "bucket-owner-full-control"
        }
      }
    },
    {
      "Sid": "AWSALBLogging2",
      "Effect": "Allow",

```

[Copy](#)

Object Ownership [Edit](#)

- Added bucket in load balancer in attributes tabs

Zone shift will be available to the load balancer.

Protection
☐ **Deletion protection**
To prevent your load balancer from being deleted accidentally, turn on deletion protection. If you turn on deletion protection, you must turn it off before you can delete the load balancer.

Monitoring
☒ **Access logs**
Access logs deliver detailed logs of all requests made to your Elastic Load Balancer. Choose an existing S3 location. If you don't specify a prefix, the logs are stored in the root of the bucket. Additional charges apply. [Learn more](#)

S3 URI
 [View](#) [Browse S3](#)
Format: s3://bucket/prefix

☐ **Connection logs**
Connection logs deliver detailed logs of all connections made to your Elastic Load Balancer. Choose an existing S3 location. If you don't specify a prefix, the logs are stored in the root of the bucket. Additional charges apply. [Learn more](#)

☐ **Health check logs - new**
Health check logs deliver detailed logs of health checks for all targets in your Elastic Load Balancer's target groups. Choose an existing S3 location. If you don't specify a prefix, the logs are stored in the root of the bucket. Additional charges apply. [Learn more](#)

[Cancel](#) [Save changes](#)

- Bucket is added in load balancer successfully

Load balancer attributes

Protection
Deletion protection: Off

Monitoring
Access logs: S3 location: [bucketforloadbalancer1](#)
Connection logs: Off
Health check logs - new: Off

- We can check load balancer logs are receiving in the s3 bucket

bucketforloadbalancer1 [Info](#)

Objects | Metadata | Properties | Permissions | Metrics | Management | Access Points

Objects (1) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create Folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 Inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

☒ Show versions [1](#) [>](#) [⚙](#)

☐ Name	☐ Type	Last modified	Size	Storage class
AWSLogs/	Folder	-	-	-